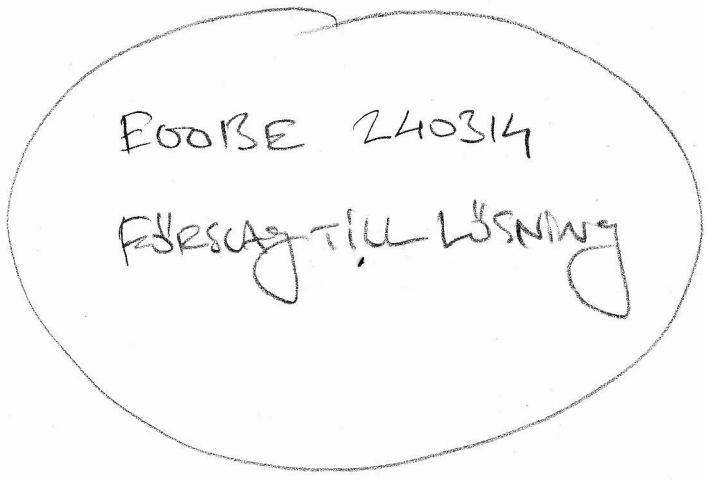
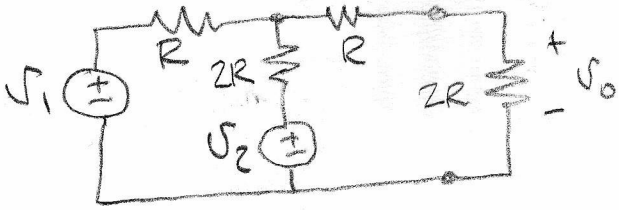
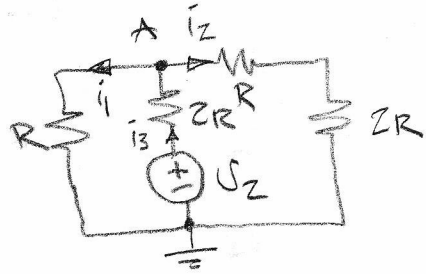

E00BE 240314
FÖRSLAG TILL LÖSNING

1.)



ANVÄNDA SUPERPOSITIONSPRINCIPEN FÖR ATT BESTÄMMA V_0

LÖSNING: BÖRJA MED ATT KORTSLUTA V_1



KSTL I NODEN GER

$$i_1 + i_2 = i_3 \quad (1)$$

DÄR

$$\begin{cases} i_1 = \frac{V_A}{R}, \; i_2 = \frac{V_A}{3R} \\ i_3 = \frac{V_2 - V_A}{2R} \end{cases} \quad (1.5)$$

VI FÅR MED (1)

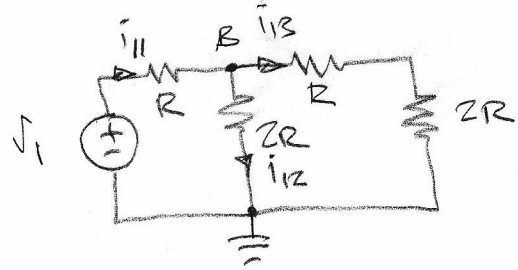
$$\frac{V_A}{R} + \frac{V_A}{3R} = \frac{V_2 - V_A}{2R}$$

$$V_A + \frac{1}{3}V_A + \frac{1}{2}V_A = \frac{1}{2}V_2$$

$$\left(\frac{6}{6} + \frac{2}{6} + \frac{3}{6}\right)V_A = \frac{1}{2}V_2$$

$$V_A = \frac{3}{11}V_2$$

KORTSLUT V_2



KSTL I NODEN GER

$$i_{11} = i_{13} + i_{12} \quad (2)$$

DÄR

$$i_{11} = \frac{V_1 - V_B}{R}$$

$$i_{12} = \frac{V_B}{2R}$$

$$i_{13} = \frac{V_B}{3R} \quad (2.5)$$

MED (2) FÅS

$$\frac{V_1 - V_B}{R} = \frac{V_B}{3R} + \frac{V_B}{2R}$$

$$V_1 = V_B + \frac{1}{3}V_B + \frac{1}{2}V_B$$

$$V_1 = \left(\frac{6}{6} + \frac{2}{6} + \frac{3}{6}\right)V_B \Rightarrow V_B = \frac{6}{11}V_1$$

↪

SPÄNNINGEN U_0 FÅS NU SOM (SUPERPONERA)

$$U_0 = 2R(i_2 + i_{13})$$

DÄR STRÖMMARNA FÅS AV (1.5) OCH (2.5)

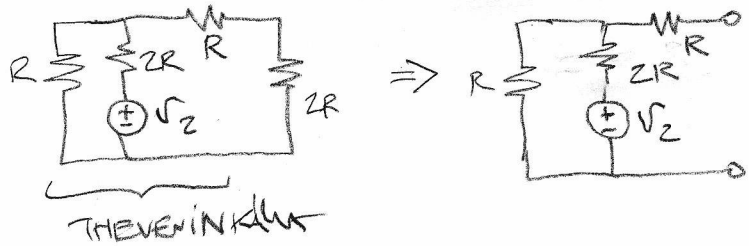
$$U_0 = 2R\left(\frac{3}{11}U_2\frac{1}{3R} + \frac{6}{11}U_1\frac{1}{3R}\right) = \frac{2}{11}(U_2 + 2U_1)$$

SVAR:

$$U_0 = \frac{2}{11}(2U_1 + U_2)$$

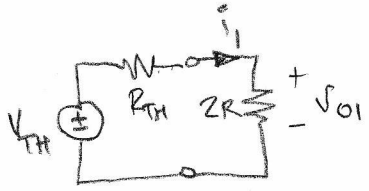
ALTERNATIV LÖSNING

(1)



$$R_{TH} = R + \left[\frac{1}{R} + \frac{1}{2R}\right]^{-1} = R + \frac{2R}{3} = \frac{5R}{3}$$

$$V_{TH} = \sqrt{2} - 2R\frac{\sqrt{2}}{3R} = \frac{1}{3}\sqrt{2}$$

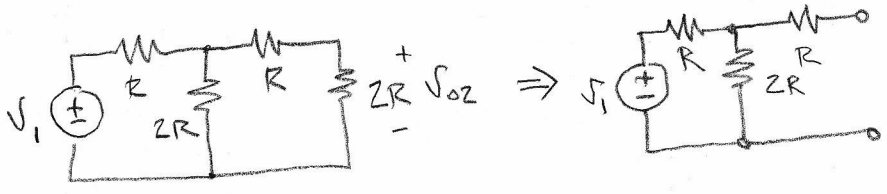


$$i_1 = \frac{V_{TH}}{R_{TH}+2R} = \frac{\frac{1}{3}\sqrt{2}}{\frac{5R}{3}+2R} = \frac{\frac{1}{3}\sqrt{2}}{\frac{5R}{3}+\frac{6R}{3}} = \frac{\frac{1}{3}\sqrt{2}}{\frac{11R}{3}} = \frac{1}{11}\sqrt{2}\frac{1}{R}$$

och spänningen

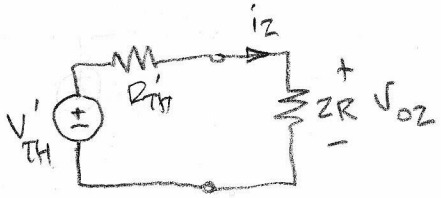
$$\sqrt{01} = 2Ri_1 = 2R\frac{1}{11}\frac{\sqrt{2}}{R} = \frac{2}{11}\sqrt{2} \quad (A)$$

(2)



$$R'_{TH} = \frac{5R}{3}$$

$$V'_{TH} = \sqrt{1} - R\frac{\sqrt{1}}{3R} = \frac{2\sqrt{1}}{3}$$



$$i_2 = \frac{V_{TH}}{R_{TH}+2R} = \frac{\frac{2\sqrt{1}}{3}}{\frac{5R}{3}+2R} = \frac{\frac{2\sqrt{1}}{3}}{\frac{11R}{3}} = \frac{2}{11}\sqrt{1}\frac{1}{R}$$

och spänningen

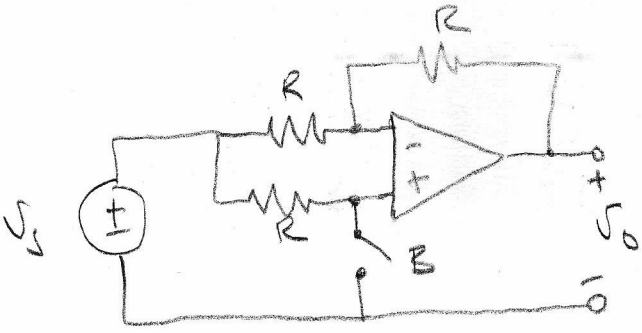
$$\sqrt{02} = 2Ri_2 = 2R\frac{2}{11}\sqrt{1}\frac{1}{R} = \frac{4}{11}\sqrt{1} \quad (B)$$

SUPERPONERA LÖSNINGARNA, (A) och (B)

$$\sqrt{0} = \sqrt{01} + \sqrt{02} = \frac{2}{11}(\sqrt{2} + 2\sqrt{1})$$

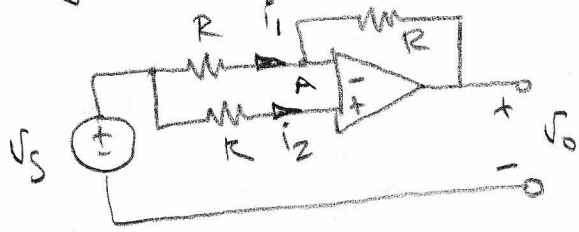
SAMMA SOM TIDIGARE.

2)



ÖPPEN, SLUTAD ?

Lösning: För B öppen gäller

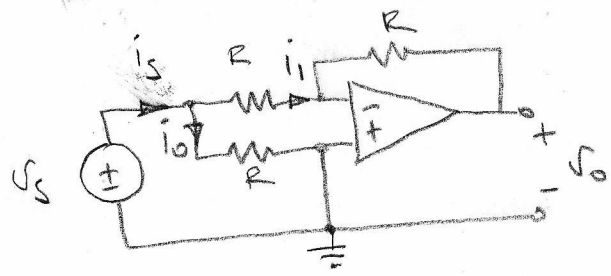


$i_2 = 0$ ty ideal OP-AMP

Vid nod A gäller att spänningen är V_s och i_1 är noll, liksom i_2. Vi får

$$i_1 = \frac{V_s - V_o}{R} = 0 \quad \text{och} \quad \underline{V_o = V_s} \qquad (1)$$

För B sluten gäller



KSPL ger $V_s - Ri_0 = 0 \Rightarrow i_0 = \frac{V_s}{R}$ (2)

$Ri_1 - Ri_0 = 0 \Rightarrow i_1 = i_0$ (3)

och $Ri_1 + V_o = 0$ (4)

(4) med (1) och (2)

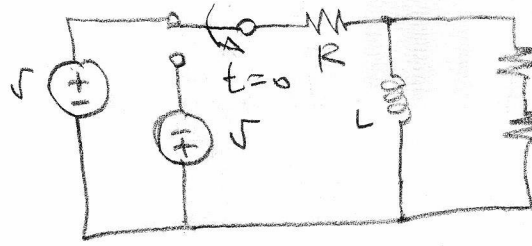
$$V_o = -R\frac{V_s}{R} = -V_s$$

ALLTSÅ $\underline{V_o = -V_s}$ PÅSTÅENDET ÄR ALLTSÅ FALSKT!

(TVÄRTOM MOT VAD SOM PÅSTÅS I FRÅGAN.)

SVAR = FALSKT.

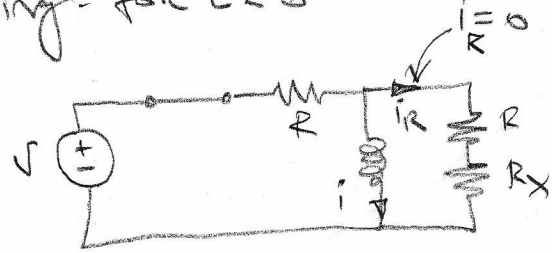
3)



$V_x(t)$?
$t \geq 0$

$V = 10V$, $R = 50\Omega$, $R_X = 75\Omega$, $L = 150\,mH$

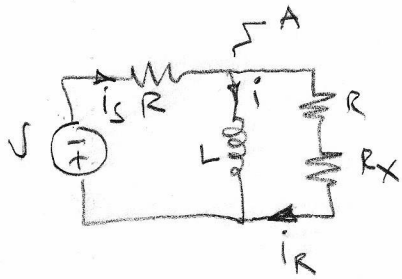
Lösning: för $t < 0$



$i = 0$ TY SPOLEN ERBJUDER INGET MOTSTÅND ALLS!

DVS $i(0) = \frac{V}{R}$ (1)

för $t \geq 0$



FÖR NOD A GÄLLER

$$i_S - i - i_R = 0 \quad (2)$$

DÄR

$$i_S = \frac{V - V_A}{R}, \quad i(t) = i(0) + \frac{1}{L}\int V_A\,dt \quad (3)$$

$$i_R = \frac{V_A}{R + R_X}$$

(3) i (2)

$$-\frac{V - V_A}{R} - i(0) - \frac{1}{L}\int V_A\,dt - \frac{V_A}{R+R_X} = 0 \quad (4)$$

DERIVERA MHPT t

$$-\frac{dV_A}{R} - \frac{1}{L}V_A - \frac{1}{R+R_X}\frac{dV_A}{dt} = 0$$

$$\underbrace{\left(\frac{1}{R} + \frac{1}{R+R_X}\right)}_{\frac{1}{\tau}}\frac{dV_A}{dt} + \frac{1}{L}V_A = 0$$

$$\frac{dV_A}{dt} + \frac{1}{\tau L}V_A = 0 \quad (5)$$

SOM HAR LÖSNING $V_A(t) = Ae^{-\frac{1}{\tau L}t}$ (6)

VAD ÄR $V_A(0)$?

STUDERA (4) för $t = 0$

(4) ger för $t=0$ $\quad = \frac{V}{R}$ enligt (1), spolen fungerar som

$$-\frac{V}{R} - \frac{V_A(0)}{R} - i(0) - 0 - \frac{V_A(0)}{R+R_X} = 0$$

$$-\frac{2V}{R} - V_A(0)\left(\frac{1}{R} + \frac{1}{R+R_X}\right) = 0$$

$$\frac{2V}{R} = V_A(0)\left(\frac{R+R_X+R}{R(R+R_X)}\right)$$

$$V_A(0) = 2V\,\frac{R+R_X}{2R+R_X} \qquad (7)$$

(7) i (6) ger

$$\begin{cases} V_A(t) = 2V\,\dfrac{R+R_X}{2R+R_X}\,e^{-\frac{1}{\tau_L}t} \\ \tau_L = \dfrac{2R+R_X}{R(R+R_X)} \end{cases}$$

Med siffror

$$V_A(t) = 2\cdot 10\cdot\frac{50+25}{100+25}\,e^{-200t} =$$

$$= 12\,e^{-200t} \text{ volt}$$

Men det var ju spänningen över R_X som sökte. Strömmen blir $i_R(t) = \dfrac{V_A}{R+R_X}$ och spänningen över R_X blir

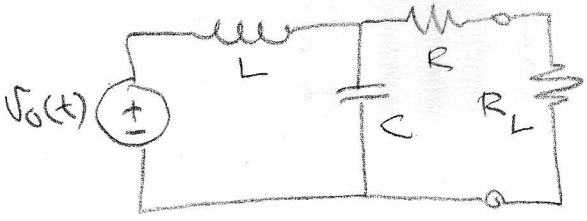
$$V_{R_X}(t) = \frac{R_X}{R+R_X}V_A(t) = \frac{25}{75}\,12\,e^{-200t} =$$

$$= 4\,e^{-200t}\ \text{V}.$$

Svar:

$$V_{R_X}(t) = 4e^{-200t}\ \text{V}$$

4.)



$V_0(t) = 30\cos(10^6 t)$

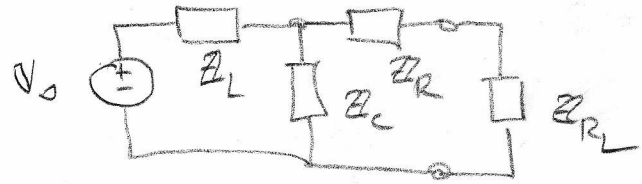
$C = 500\,pF$

$R_L = 500\,\Omega$

$R = 1000\,\Omega$

$L = 1\,MH$

Lösning: Komplex Representation



DÅ R

$Z_L = j\omega L = j1000\,\Omega$

$Z_C = \frac{1}{j\omega C} = -j2000\,\Omega$

$Z_R = 1000\,\Omega$

$Z_{R_L} = 500\,\Omega$

EKV. IMPEDANS

$$Z_{EK} = Z_L + \frac{Z_C(Z_R + Z_{R_L})}{Z_C + Z_R + Z_{R_L}} =$$

$$= 10^3 j + \frac{-2000j \cdot 1500}{-2000j + 1500} = 10^3 j + \frac{-3\cdot 10^6 j}{2500\angle -53.13^\circ} =$$

$$= 10^3 j + 1200\angle -36.87^\circ \simeq 960 + 280j =$$

$$= 1000\angle 16.26^\circ\ \Omega$$

STRÖMMEN BLIR

$$I_0 = \frac{V_0}{Z_0} = \frac{30\angle 0^\circ}{1000\angle 16.26^\circ} = 30\angle -16.26^\circ\ mA$$

Spänningen i NOD A. $V_A = V_0 - Z_L I_0$.

$$\Rightarrow V_A = 30\angle 0^\circ - 1\cdot 10^3 j\, 30\cdot 10^{-3}\angle 16.26^\circ =$$

$$= 30\angle 0^\circ - 30\angle 90^\circ - 16.26^\circ =$$

$$= 30 - 8.4 - 28.80j = 21.6 - 28.8j =$$

$$= 36\angle -53.13^\circ \text{ VOLT.}$$

STRÖMMEN genom högra delen, I_x, BLIR

$$I_x = \frac{V_A}{R+R_L}$$

KOMPLEX EFFEKT SOM R_L FÖRBRUKAR BLIR

$$\$_{R_L} = \frac{1}{2} V_{R_L} I_x^* = \frac{1}{2} Z_{R_L} I_x I_x^* = \underline{\frac{1}{2} Z_{R_L} |I_x|^2}$$

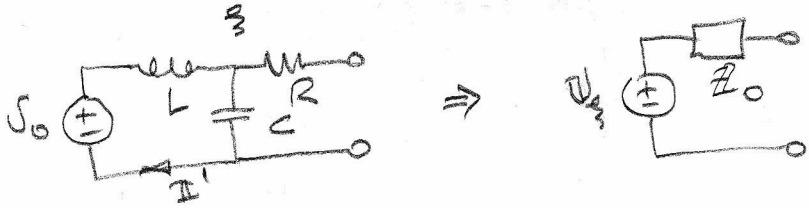
Vi FÅR

$$\$_{R_L} = \frac{1}{2} 500 \left(\frac{36}{1500}\right)^2 = \underline{0.144 \text{ VA}}$$

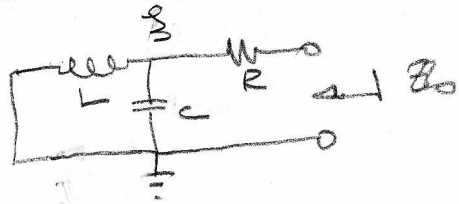
$\underline{\text{SVAR}}$: $\$_{R_L} = \underline{144 \text{ mVA}}$

ALTERNATIV LÖSNING

GÖR EN THEVENINEKVIVALENT.



FÖRST. IMPEDANSEN Z_0



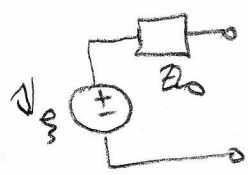
$$Z_0 = R + \frac{j\omega L \cdot \frac{1}{j\omega C}}{\frac{1}{j\omega C} + j\omega L} = R + \frac{\frac{L}{C} j\omega C}{1 - \omega^2 LC} = \frac{R(1-\omega^2 LC) + j\omega L}{1-\omega^2 LC} \quad (A)$$

SEDAN SPÄNNINGEN I NOD ξ

$$\begin{cases} V_0 - \left(j\omega L + \frac{1}{j\omega C}\right) I' = 0 \Rightarrow I' = \dfrac{V_0}{j\omega L + \frac{1}{j\omega C}} & (B) \\ V_\xi = \frac{1}{j\omega C} I' & (C) \end{cases}$$

$$(B) \text{ i } (C) \quad V_\xi = \frac{1}{j\omega C} \frac{V_0}{j\omega L + \frac{1}{j\omega C}} = \frac{V_0}{1-\omega^2 LC} \quad (D)$$

THEVENINKRETSEN BLIR ALLTSÅ



DÄR $\begin{cases} V_\xi = \dfrac{V_0}{1-\omega^2 LC} \\ Z_0 = \dfrac{R(1-\omega^2 LC) + j\omega L}{1-\omega^2 LC} \end{cases}$

KOMPLEX EFFEKT SOM R_L FÖRBRUKAR BLIR

$$S_{R_L} = \frac{1}{2} V_{R_L} I_{R_L}^* = \frac{1}{2} Z_{R_L} I_{R_L} I_{R_L}^* = \frac{1}{2} Z_{R_L} |I_{R_L}|^2$$

DÄR

$$I_{R_L} = \frac{V_\xi}{Z_0 + Z_{R_L}} = \frac{\frac{V_0}{1-\omega^2 LC} \cdot (1-\omega^2 LC)}{R(1-\omega^2 LC) + j\omega L + Z_{R_L}(1-\omega^2 LC)} =$$

$$= \frac{V_0}{(R+R_L)(1-\omega^2 LC) + j\omega L} \quad \hookrightarrow$$

MED SIFFROR

$$\underline{I}_{RL} = \frac{30\angle 0^\circ}{1500\left(1-10^{12}\cdot 1\cdot 10^{-3}\cdot 500\cdot 10^{-12}\right)+j10^{6}\cdot 1\cdot 10^{-3}} =$$

$$= \frac{30\angle 0^\circ}{750+j1000} = \frac{30\angle 0^\circ}{1250\angle 53.13^\circ} =$$

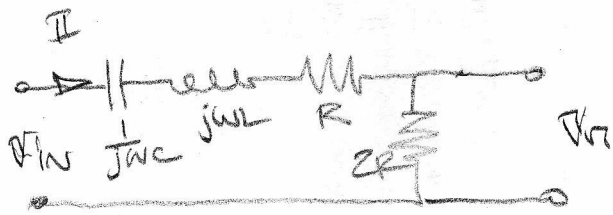
$$= 0.024\angle -53.13^\circ$$

OCH KOMPLEXA EFFEKTEN

$$\underline{S}_{RL} = \tfrac{1}{2}\,500\,(0.024)^2 = 0.144 \text{ VA}$$

... SAMMA SOM FÖRUT.

5)



a) Rescim $H(\omega)$. KSPL ecuatii

$$\begin{cases} -V_{IN} + \left(\frac{1}{j\omega C} + j\omega L + 3R\right) I = 0 \\ V_{UT} - 2RI = 0 \end{cases}$$

$$\Rightarrow V_{IN} + \left(\frac{1}{j\omega C} + j\omega L + 3R\right)\frac{V_{UT}}{2R} = 0$$

$$\frac{V_{UT}}{V_{IN}} = \frac{2R}{\frac{1}{j\omega C} + j\omega L + 2R} = \frac{2Rj\omega C}{1 - \omega^2 CL + 3j\omega CR} \quad (1)$$

$$H(\omega) = \frac{2Rj\omega C}{1 - \omega^2 CL + 3j\omega CR}$$

Rescriem $Q(\omega)$

VR (1) FAS

$$H(\omega) = \frac{2Rj\omega C\left((1-\omega^2 CL) - 3j\omega RC\right)}{(1-\omega^2 CL)^2 + (3\omega CR)^2} =$$

$$= \frac{2Rj\omega C(1-\omega^2 CL) + 6(\omega RC)^2}{(1-\omega^2 CL)^2 + (3\omega CR)^2} \quad (2)$$

VR (2)

$$Q(\omega) = \text{ATAN}\left(\frac{Im(H(\omega))}{Re(H(\omega))}\right) = \text{ATAN}\left(\frac{2R\omega C(1-\omega^2 CL)}{6(\omega RC)^2}\right) =$$

$$= \text{ATAN}\left(\frac{1}{3}\,\frac{1-\omega^2 CL}{\omega RC}\right)$$



b) För vilket ω är V_R och V_{in} i fas?

I fas om $H(\omega)$ är reell, dvs $\mathrm{Im}(H(\omega)) = 0$

$$\Rightarrow \mathrm{Im}(H(\omega)) = \frac{2RC\omega(1-\omega^2LC)}{(1-\omega^2LC)^2 + (3\omega CR)^2} = 0$$

$$\Rightarrow 1 - \omega_0^2 LC = 0$$

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

För den vinkelfrekvensen blir

$$|H(\omega_0)| = \frac{6(\omega_0 RC)^2}{(3\omega_0 CR)^2} = \frac{6}{9} = \frac{2}{3}$$

och

$$\varphi(\omega_0) = 0°$$

SVAR: a) $H(\omega) = \dfrac{2RC\omega}{1-\omega^2LC+3j\omega RC}$

$$\varphi(\omega) = \mathrm{ATAN}\left(\frac{1}{3}\,\frac{1-\omega^2LC}{\omega RC}\right)$$

b) $\omega_0 = \dfrac{1}{\sqrt{LC}}$

$|H(\omega_0)| = \frac{2}{3}$

$\varphi(\omega_0) = 0°$